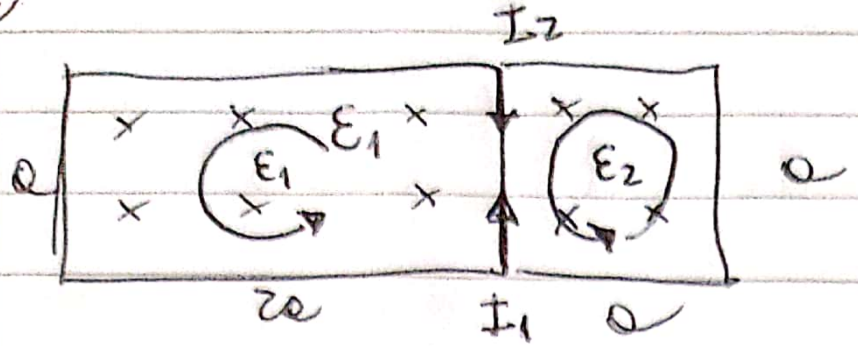
3



$\boxed{B \text{ aumenta en } t}$

ϕ aumenta con "t"

$A_1 = 2a^2$

$$\phi = \int B\,dA$$

$$= BA$$

$$= 0{,}001t \cdot 2a^2$$

$A_2 = a^2$

$$\phi = 0{,}001t \cdot a^2$$

Escaneado con CamScanner

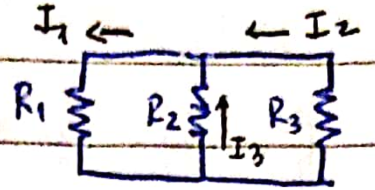
$l_1 = 5a$ $\qquad$ $l_3 = a$

$l_2 = 3a$

$\varepsilon = -d\phi/dt \Rightarrow A_1 \Rightarrow \varepsilon_1 = -2 \cdot 0{,}001 \cdot a^2$

$R_i = 0{,}1 \cdot l_i$ $\qquad$ $= -0{,}845\ [mV]$.

$A_2 \Rightarrow \varepsilon_2 = -0{,}423\ [mV]$.



$R_1 = 0{,}1 \cdot (5 \cdot 0{,}65)$

$R_2 = 0{,}1 \cdot (3 \cdot 0{,}65)$

$R_3 = 0{,}1 \cdot (0{,}65)$

$R_1 = 0{,}1 \cdot \frac{\Omega}{m} \cdot 5 \cdot 0{,}65\ m = 0{,}325\ \Omega$

$R_2 = 0{,}1 \cdot 3 \cdot 0{,}65 = 0{,}195\ \Omega$

$R_3 = 0{,}1 \cdot 0{,}65 = 0{,}065\ \Omega$

$|\varepsilon_1| - I_1 R_1 - I_3 R_2 = 0$; $\quad 0{,}845 \cdot 10^{-3} - 0{,}325 I_1 - 0{,}195 I_3 = 0$

$|\varepsilon_2| + I_3 R_2 - I_2 R_3 = 0$; $\quad 0{,}423 \cdot 10^{-3} + 0{,}195 I_3 - 0{,}065 I_2 = 0$

$I_3 + I_2 = I_1$ $\qquad$ $I_3 + I_2 = I_1$

$I_1 = 3{,}11\ [mA]$

$I_2 = 3{,}96\ [mA]$

$I_3 = -0{,}85\ [mA]$ $\qquad \downarrow I_3$